(PV-SDS-01) SwSDS

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Revision History

Rev.	Date	Description
`0.0`	`2012.07.02`	Initial version
`0.1`	`2015.01.12`	User interface update
`0.2`	`2016.01.19`	New function implemented
`0.3`	`2017.01.13`	System issue revision

Rev.	Date	Description
`0.4`	`2018.01.12`	New device added
`0.5`	`2019.01.21`	Device compatibility update
`0.6`	`2020.01.30`	Device compatibility update
`0.7`	`2020.10.08`	GUI update
`0.8`	`2021.02.26`	Linkage usability improvement
`0.9`	`2021.09.10`	Program issue update
`1.0`	`2021.10.08`	Device compatibility update
`1.1`	`2022.03.04`	Connection-status improvement
`1.2`	`2023.01.04`	Increased image capacity
`1.3`	`2023.04.21`	Device compatibility update
`1.4`	`2024.01.08`	Device compatibility update
`1.5`	`2024.01.15`	Document number changed from 603 to Z01 according to OP-709
`1.6`	`2025.08.14`	Updated software architecture and added translation-related design item

1. Purpose

This document describes the software design needed to satisfy the Portview software requirements. It specifies the software architecture, data design, and detailed design of each software unit to a level sufficient for implementation and verification.

2. Design Overview

Portview is designed as workstation software for image management, image acquisition, image viewing, and supporting diagnostic workflow operations.

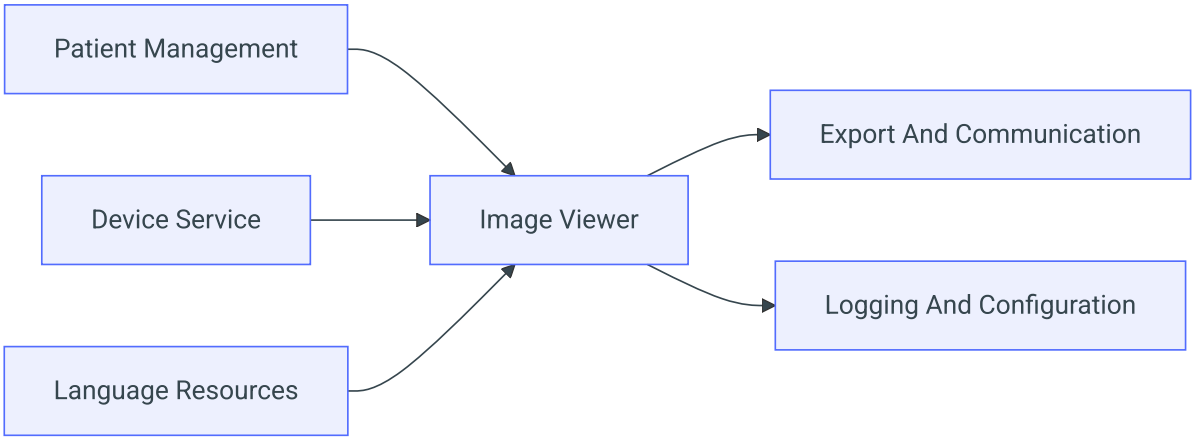
The software is integrated with an image database, image viewer, and image acquisition module. The design separates the software into stable operational units:

- patient management
- device service
- image viewer
- export and communication services
- language-resource support

ImageProcess-specific design content is not carried into the current authored scope.

3. System Architecture

3.1 Software Architecture



- 1) Patient management unit manages patients' information (add / delete / modify).
- 2) Device service unit checks connectivity between Viewer and modalities.
- 3) Image viewer unit shows images relevant to the selected patient and supports annotations (length, arrow, ellipse, angle).
- 4) Exportation service unit exports images to other types (DICOM files, plain images, burned CD).
- 5) Image viewer should offer GUI translations when the user demands.

3.2 Module Responsibilities

Module	Responsibility
Patient management unit	Manage patient information including add, delete, and modify operations

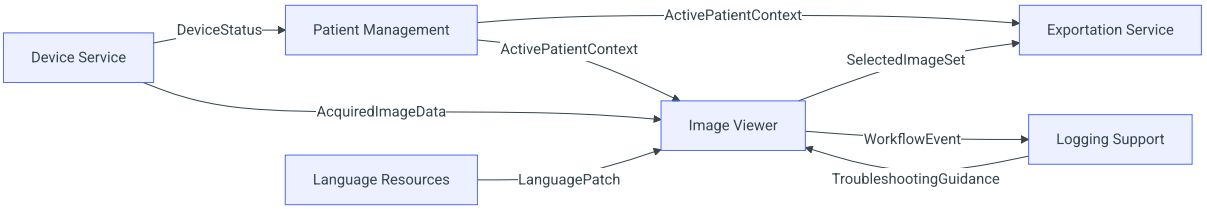
Module	Responsibility
Device service unit	Check connectivity between the viewer and connected modalities
Image viewer unit	Display images and provide diagnostic interaction tools
Exportation service unit	Export images and related information to supported destinations
Language-resource support	Provide GUI translation support

3.3 Representative Processing Paths

Processing Path	Design Intent	Primary Modules
Patient registration and selection	Create, update, select, and reopen patient or study records before image operations start	Patient management, local data store
Acquisition and import	Acquire images from supported sensors, file imports, or TWAIN paths and bind them to the active patient context	Device service, patient management, image viewer
Mount and viewer reconstruction	Reconstruct the correct mount, image set, and annotation set from stored metadata and file structures	Image viewer, TMI/TII files, local data store
Annotation and diagnostic interaction	Create and render annotation objects while preserving image and patient attribution	Image viewer, annotation modules
Output and communication	Print, export, burn media, or transmit DICOM payloads while preserving selected-image context	Exportation service, image viewer
Logging and support	Record operational events, support troubleshooting, and apply language-resource updates	Logging support, language-resource support

3.4 Inter-Unit Interface Definitions

Source Unit	Target Unit	Transferred Data	Trigger	Failure Handling
Patient Management	Image Viewer	`ActivePatientContext` (PatientID, StudyGUID, EquipmentPath)	Patient selection or reopen completed	Viewer receives empty context; display error message
Patient Management	Exportation Service	`ActivePatientContext` (PatientID, StudyGUID)	Export action initiated by user	Export aborted; user-visible error
Device Service	Image Viewer	`AcquiredImageData` (image buffer, modality metadata, patient binding)	Acquisition completed	Viewer not updated; acquisition error displayed
Device Service	Patient Management	`DeviceStatus` (connection state, device identifier)	Device state change detected	Status display updated to disconnected state
Image Viewer	Exportation Service	`SelectedImageSet` (image references, current annotations, display metadata)	User initiates print, export, or DICOM send	Export aborted if no image selected; error displayed
Image Viewer	Logging Support	`WorkflowEvent` (event type, source module, timestamp)	Any auditable viewer operation	Log failure does not block primary workflow
Language Resources	Image Viewer	`LanguagePatch` (resource key-value set, target locale)	User changes GUI language	GUI falls back to default language on patch failure
Logging Support	Image Viewer	`Troubleshooting Guidance` (error type, instruction text)	System instability or classified error detected	Generic error message if guidance lookup fails



4. Data Design

4.1 Database Study Table

Data Group	Representative Fields	Design Use
Record identity	`IdxStudy`, `StudyInstanceUID`, `GUID`, `BackupGUID`	Stable identity for patient-study records and backup linkage
Patient identity	`ID`, `OtherID`, `Name`, `BirthDate`, `Sex`, `Age`	Patient registration, lookup, and demographic display
Clinical and workflow context	`StudyDescription`, `BodyPartExamined`, `PerfPrcStepID`, `AccessNumber`	Study meaning and workflow attribution
State and storage	`Opened`, `BackupStatus`, `WhereIs`, `BackupHDDID`	Reopen behavior, backup tracking, storage-state management
Counts and category	`CountPX`, `CountCX`, `CountOX`, `CountOV`, `CountDC`, `CountCT`, `category`	Acquisition composition and mounted-study organization
Contact and address	`Email`, `Address1`, `City`, `StateProvince`, `HomePhone`, `MobilePhone`	Patient-contact and administrative data

4.2 Non-Database Files

File Structure	Purpose	Representative Fields
`TMI` (FILM_MOUNT_INFO)	Mounted-image layout	`TYPE`, `GUID`, `WIDTH`, `HEIGHT`, `DIVIDE_COL`, `DIVIDE_ROW`, `IMAGE_FILE` (148 bytes)

File Structure	Purpose	Representative Fields
`TMI` (ITEM)	Mount cell placement	`LEFT`, `TOP`, `RIGHT`, `BOTTOM`, `GUID`, `IMAGE_FILES`, `SELECTED_IMAGE`, `TITLE` (528 bytes)
`TII` (IMAGE_INFO)	Extra image metadata	`nImageType`, `nWidth`, `nHeight`, `nUsingBits`, `nInverted`, `nLaterality`, `Comment`, `dCliFactor`, `nW1OffsetToDisp`, `nW2OffsetToDisp` (2038 bytes)

4.3 Persistence And Reconstruction Logic

The viewer reconstructs the active display state from:

- database-managed patient and study records
- mount-layout metadata in `TMI`
- image and calibration metadata in `TII`
- annotation records linked to the selected equipment or image path

5. Detailed Design Specifications

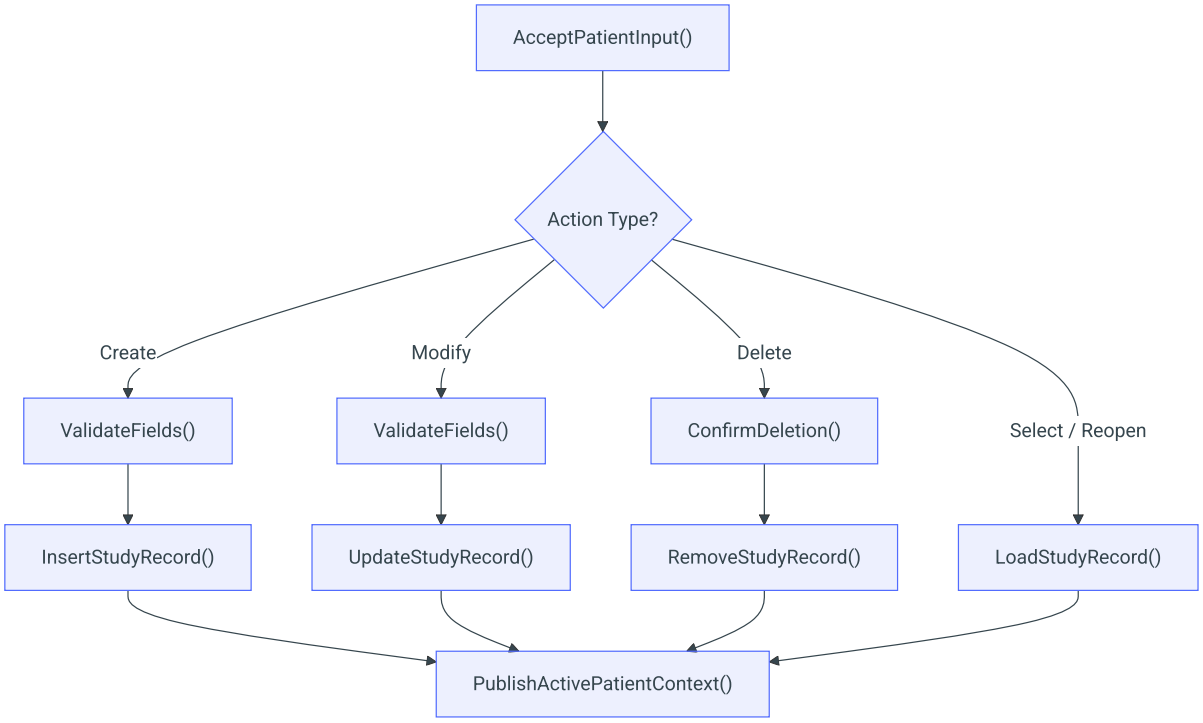
5.0 Design Allocation Summary

Design Area	Design Items
Patient and study handling	`SDS-001`
Acquisition and device interaction	`SDS-002`, `SDS-023` to `SDS-030`
Viewer presentation	`SDS-003`, `SDS-004`, `SDS-005`, `SDS-006`
Image interaction	`SDS-013`, `SDS-014`
Annotation and measurement	`SDS-007` to `SDS-012`
Output and communication	`SDS-015` to `SDS-019`
Support functions	`SDS-020`, `SDS-021`, `SDS-022`, `SDS-031`

5.1 Patient And Study Handling

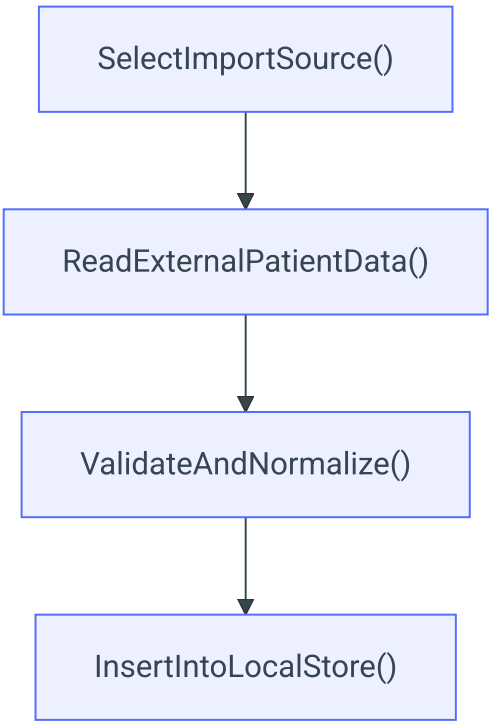
SDS-001: Patient Management

Aspect	Detail
Input	Patient information (demographics, study metadata, user actions)
Output	Persistent patient-study records, active patient context
Algorithm	Validate and persist patient data, maintain active record context, expose correct study to downstream modules



Sub-process: Patient Management (Import)

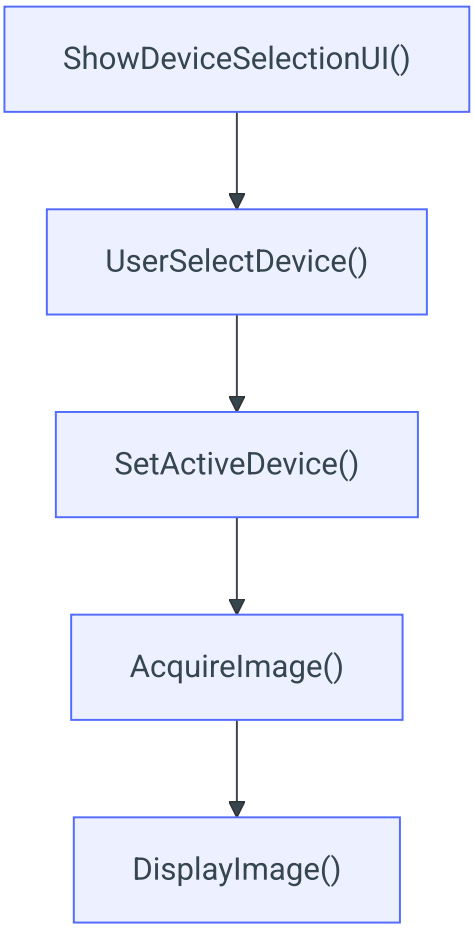
Aspect	Detail
Input	External patient data from disc or file
Output	Imported patient record bound to local data store



5.2 Acquisition And Device Interaction

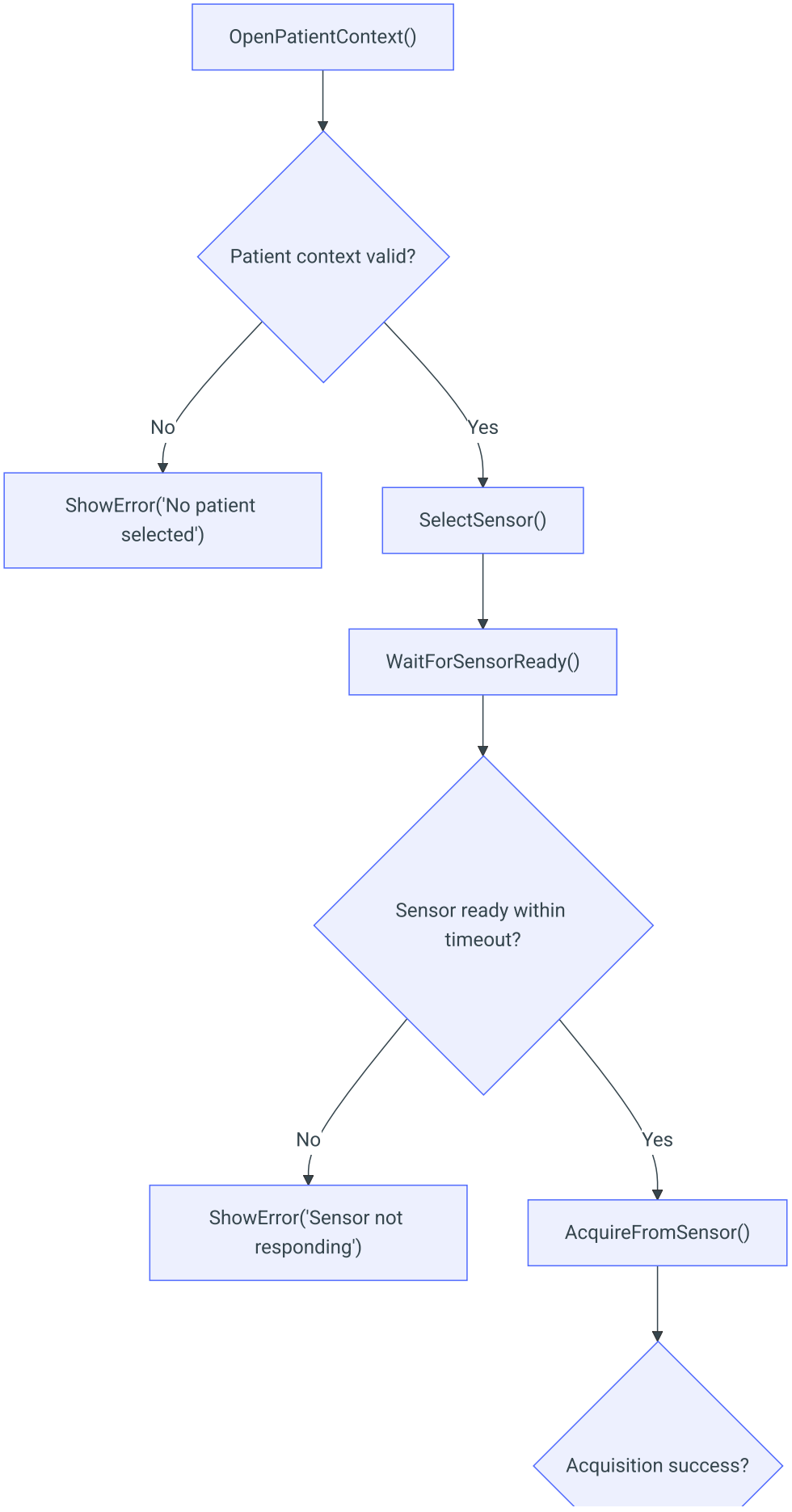
SDS-023: Select Device And Acquire Image

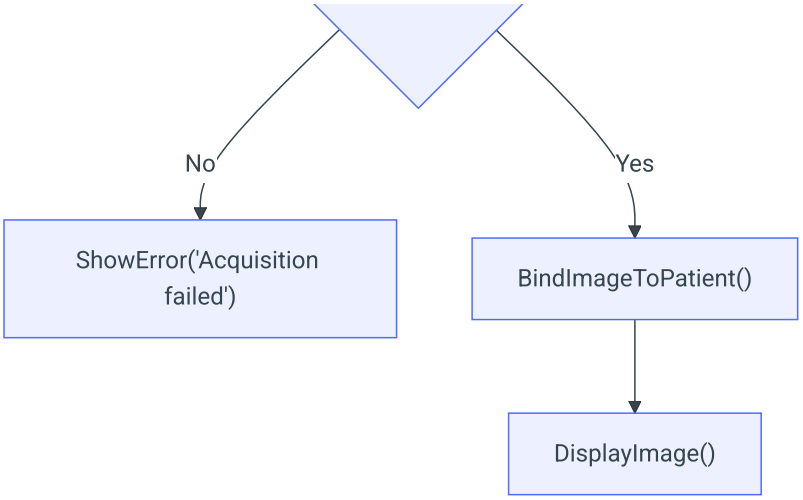
Aspect	Detail
Input	Available device list, user device selection
Output	Active device path, acquired image data



SDS-024: Acquire Image From Intraoral Sensor

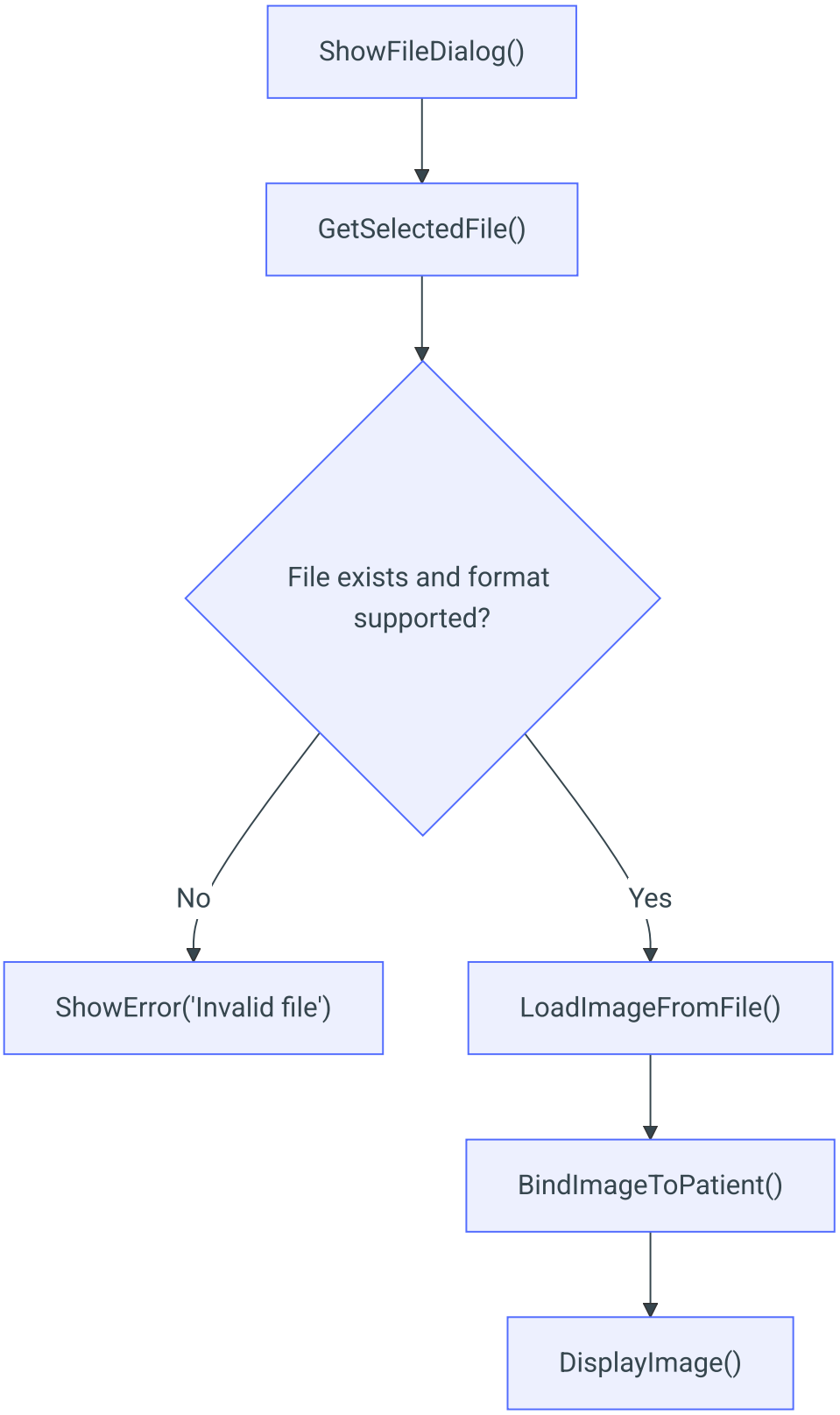
Aspect	Detail
Input	Intraoral sensor signal, selected patient context
Output	Image data bound to active patient
Error handling	Sensor timeout, sensor not responding, patient context missing


OpenPatientContext()
Patient context valid?
No
Yes
ShowError('No patient selected')
SelectSensor()
WaitForSensorReady()
Sensor ready within timeout?
No
Yes
ShowError('Sensor not responding')
AcquireFromSensor()
Acquisition success?



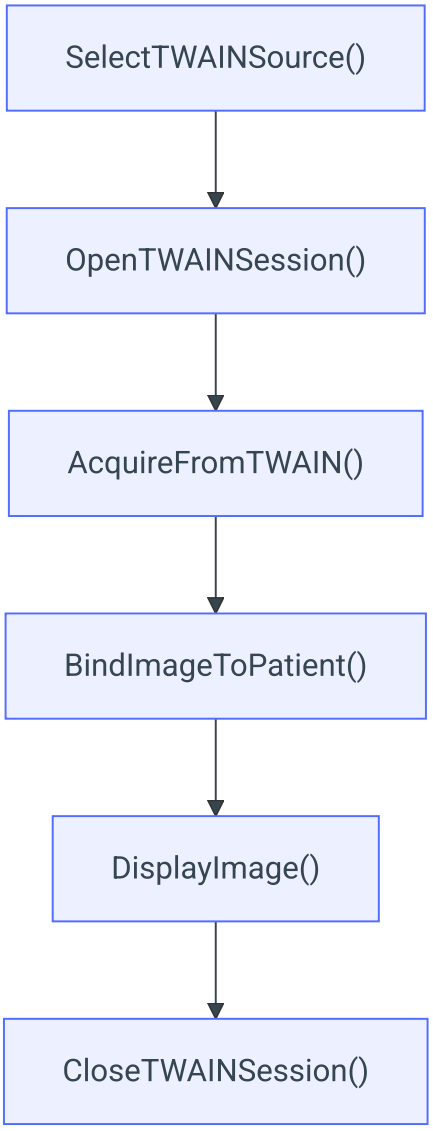
SDS-002: Acquire Image From File

Aspect	Detail
Input	File path
Output	Image bound to active patient
Error handling	File not found, unsupported format, patient context missing



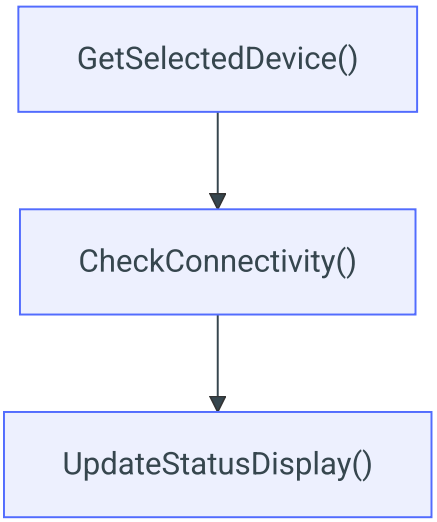
SDS-025: Acquire Image From TWAIN Device

Aspect	Detail
Input	TWAIN device signal
Output	Image bound to active patient



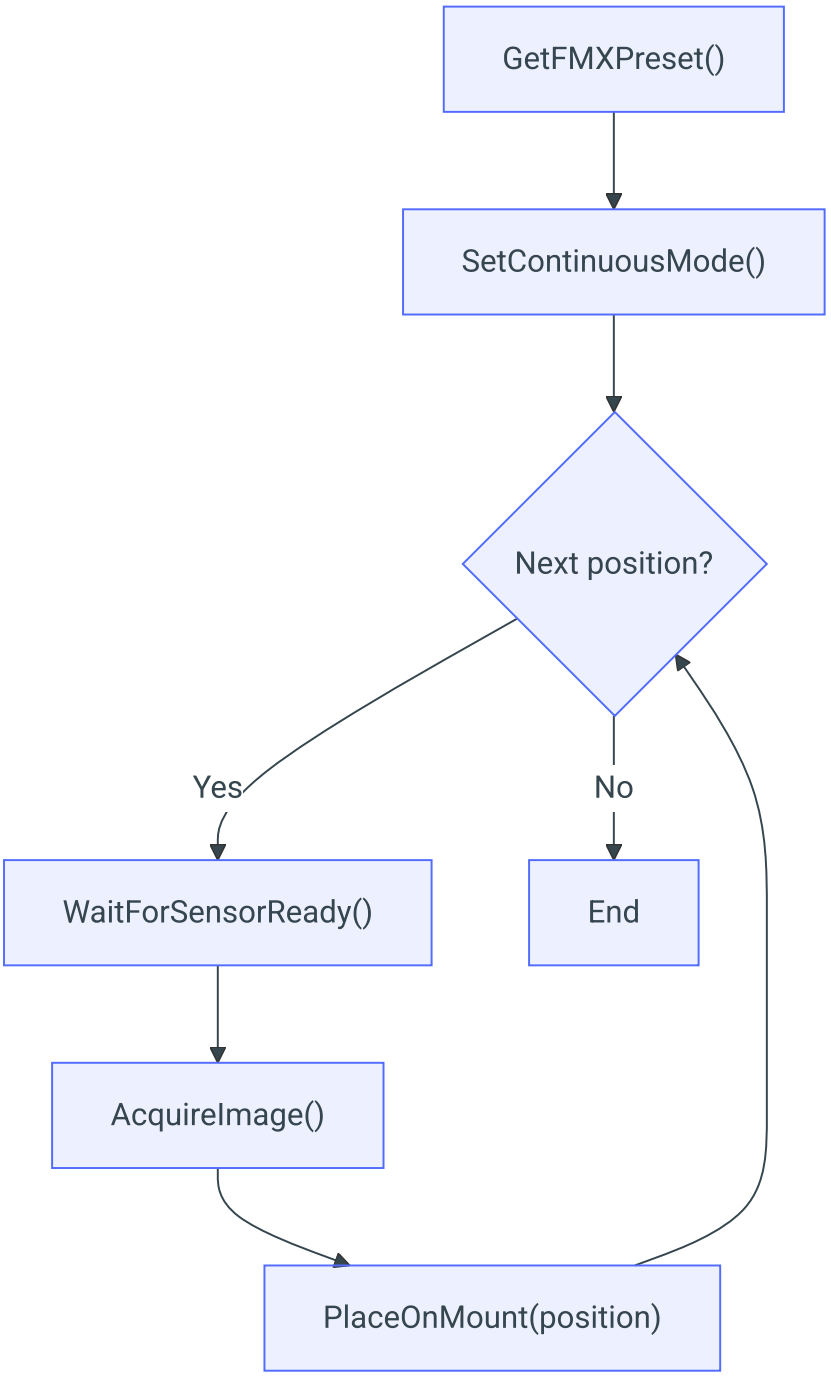
SDS-026: Show Status Of Device

Aspect	Detail
Input	Selected device, connection state
Output	Device status display



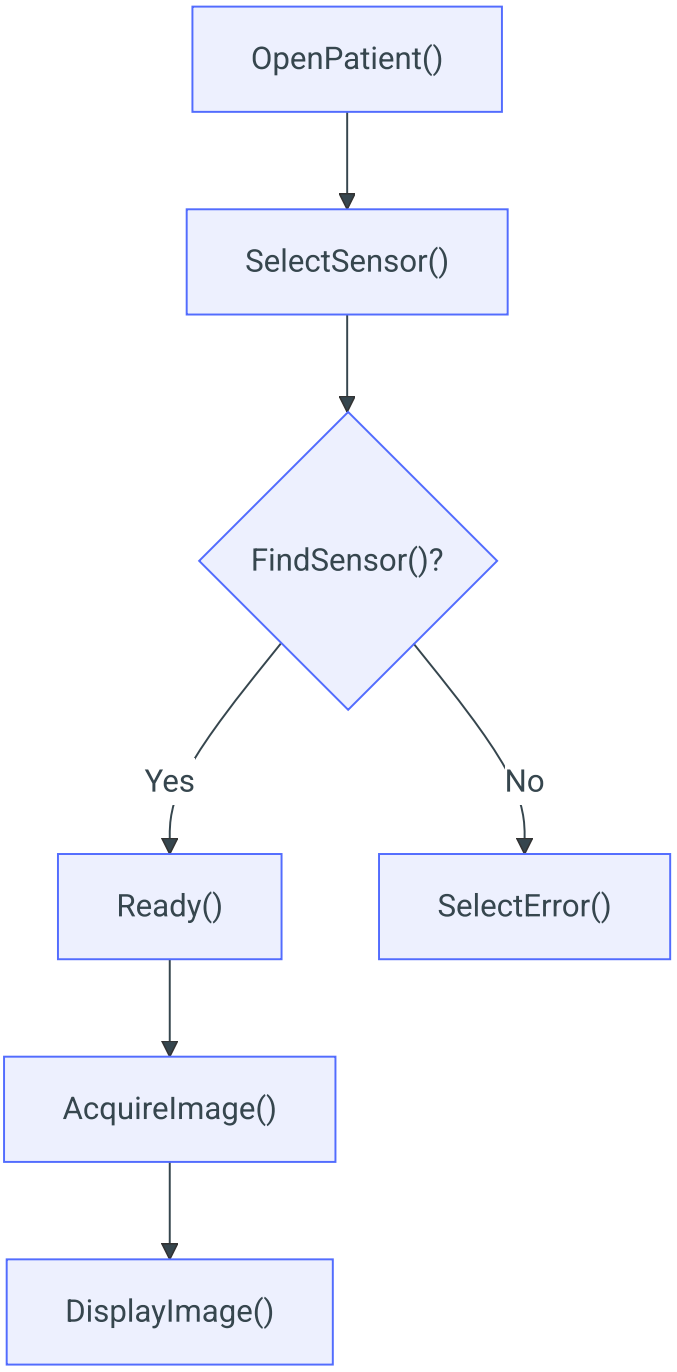
SDS-027: Change FMX Continuous Acquisition Mode

Aspect	Detail
Input	Preset order, FMX layout
Output	Sequential acquisition workflow



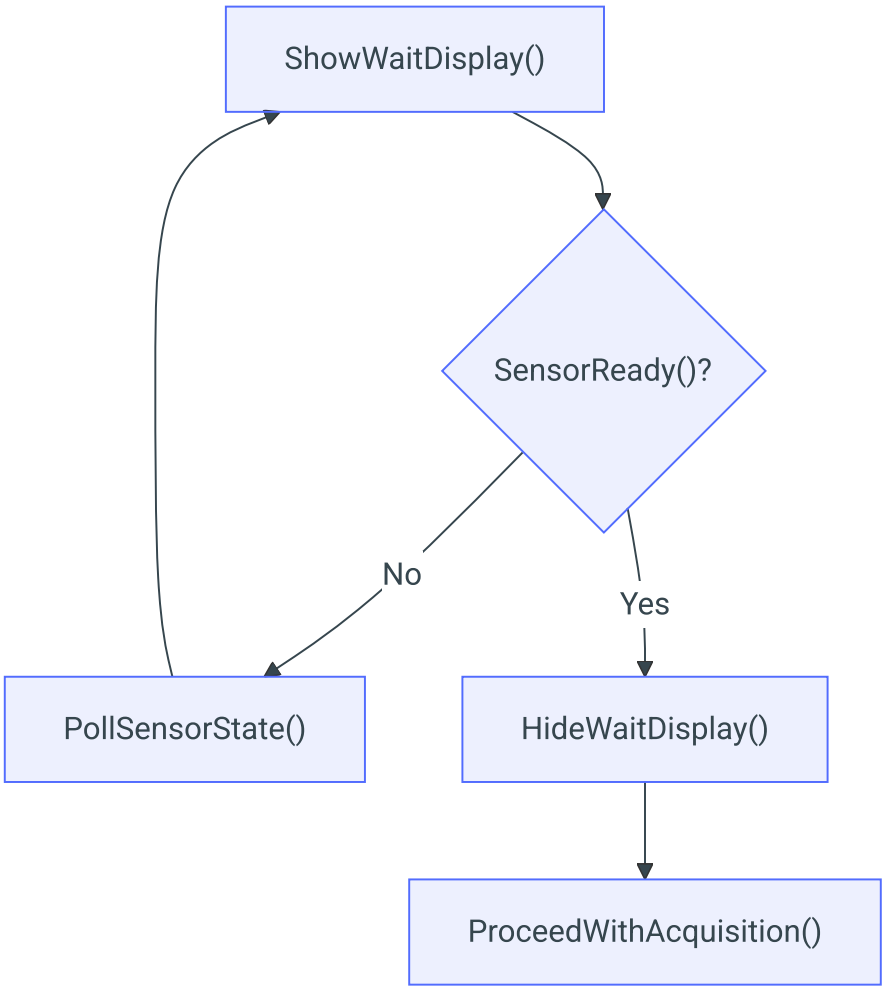
SDS-028: Multi-Sensor Select

Aspect	Detail
Input	Sensor connection signals
Output	Acquisition image from selected sensor



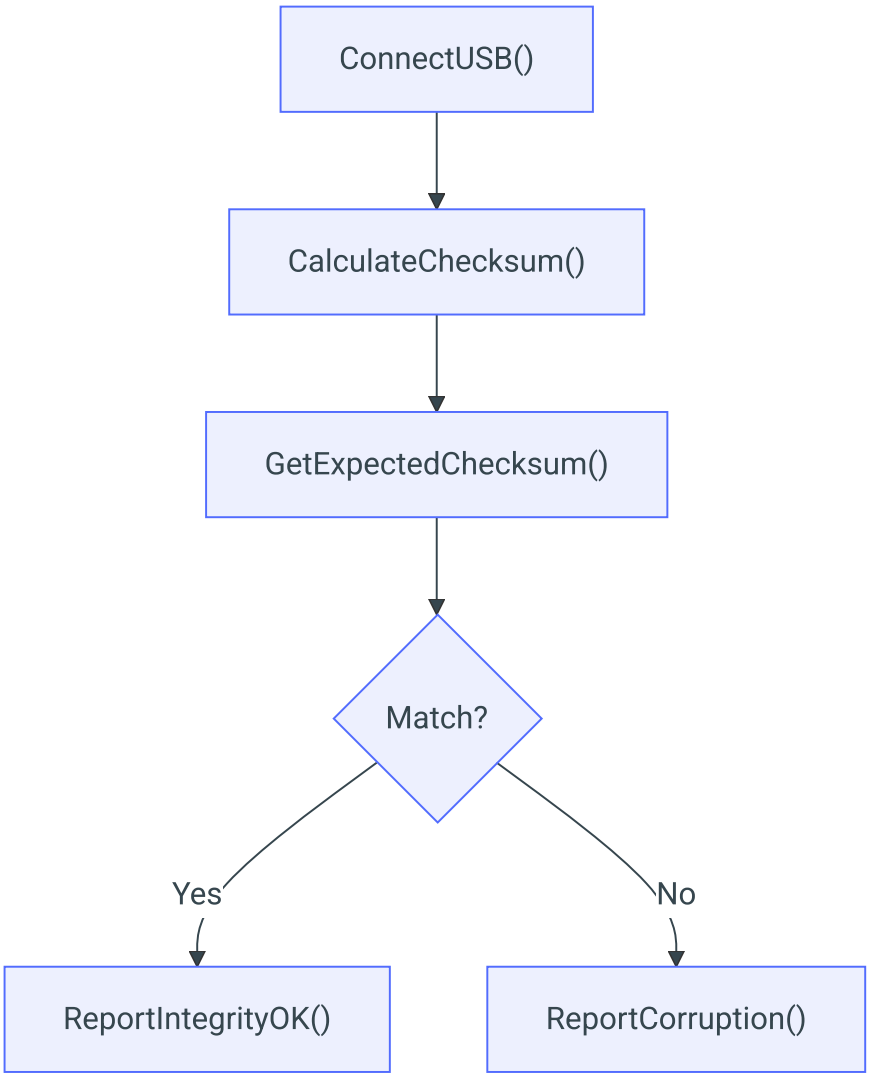
SDS-029: Intraoral Sensor Acquisition Wait Display

Aspect	Detail
Input	Sensor readiness state
Output	Wait-state display to user



SDS-030: USB Data Integrity Check

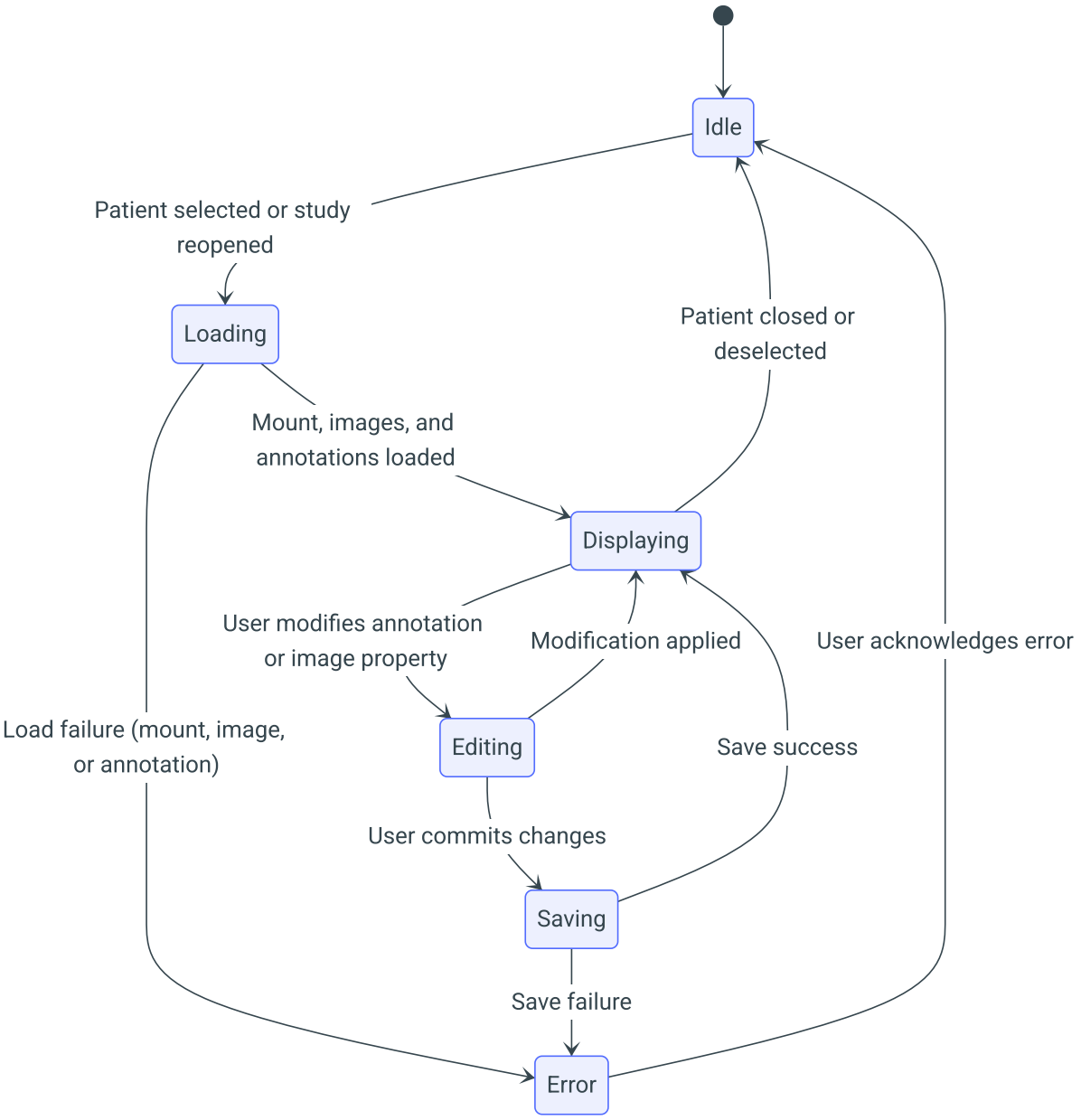
Aspect	Detail
Input	USB drive
Output	Result of the integrity check



5.3 Viewer Presentation

Viewer State Model

The Main Viewer operates through the following state transitions:

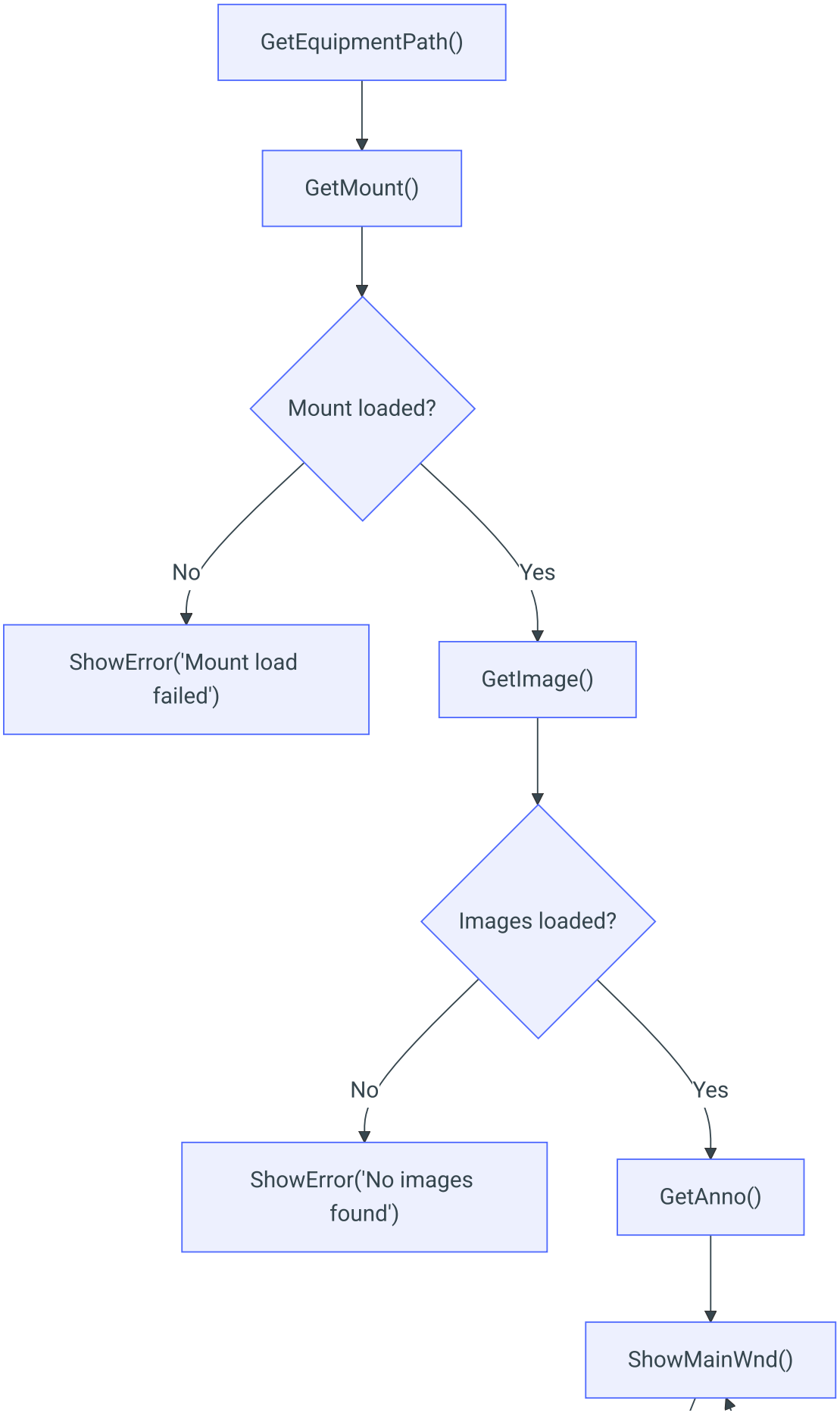


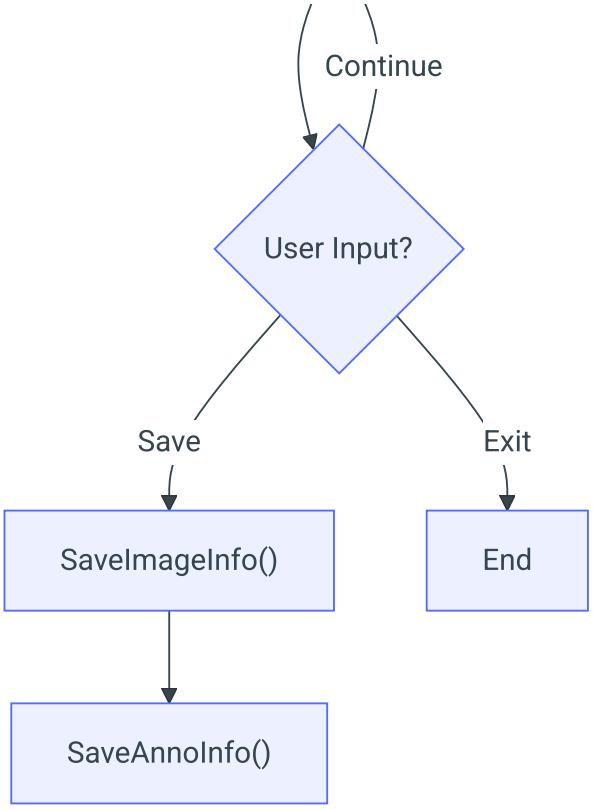
State	Description
Idle	No patient context active; viewer is empty
Loading	Reconstructing mount, images, and annotations from stored metadata
Displaying	Active patient images and annotations rendered on mount
Editing	User is modifying annotation geometry, text, or image display properties

State	Description
Saving	Committing modified image info and annotation info to persistent storage
Error	A load or save failure has occurred; user action required to recover

SDS-003: Main Viewer

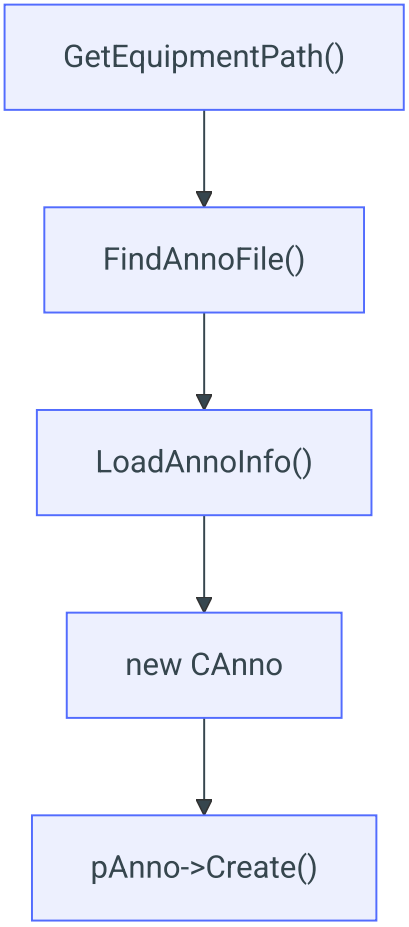
Aspect	Detail
Input	Patient image path, mount metadata, annotation metadata
Output	Display images and annotations on mount
Error handling	Mount load failure, image load failure, annotation load failure, save failure


GetEquipmentPath()
GetMount()
Mount loaded?
No
Yes
ShowError('Mount load failed')
GetImage()
Images loaded?
No
Yes
ShowError('No images found')
GetAnno()
ShowMainWnd()



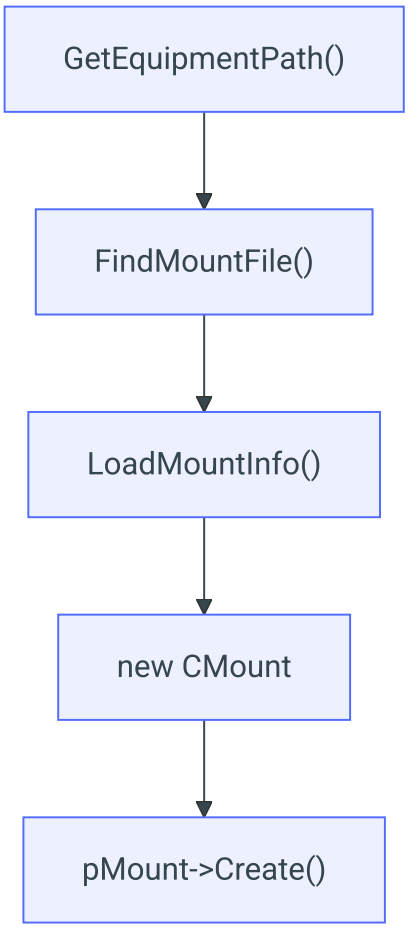
Sub-process: GetAnno

Aspect	Detail
Input	Equipment path
Output	Created annotation objects



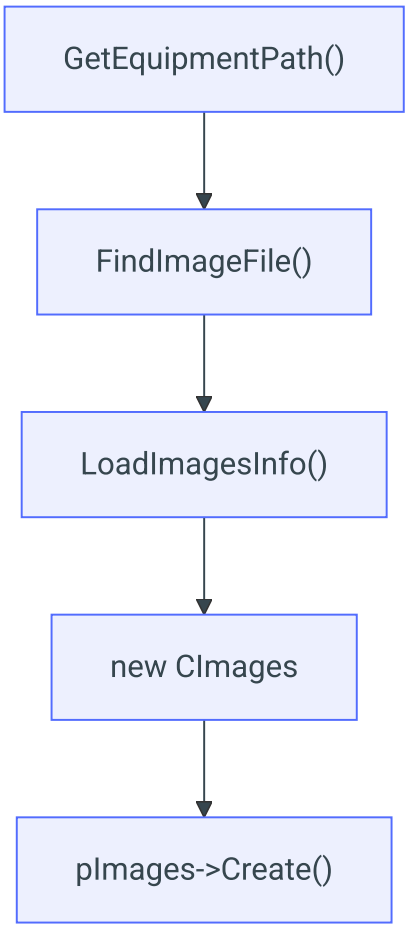
Sub-process: GetMount

Aspect	Detail
Input	Equipment path
Output	Created mount template



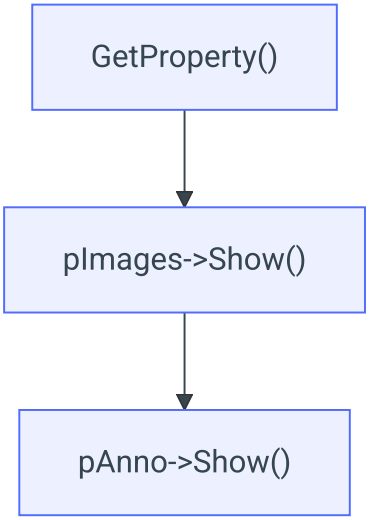
Sub-process: GetImage

Aspect	Detail
Input	Equipment path
Output	Created image objects



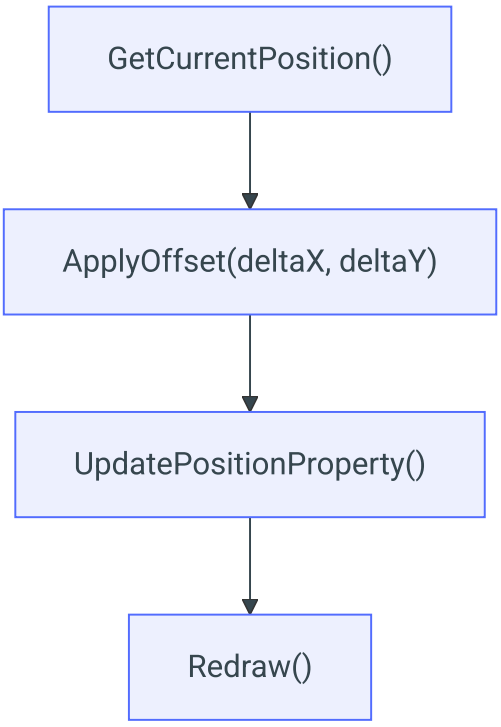
Sub-process: ShowMainWnd / ShowMainWndImages

Aspect	Detail
Input	Equipment type properties
Output	Rendered images and annotations



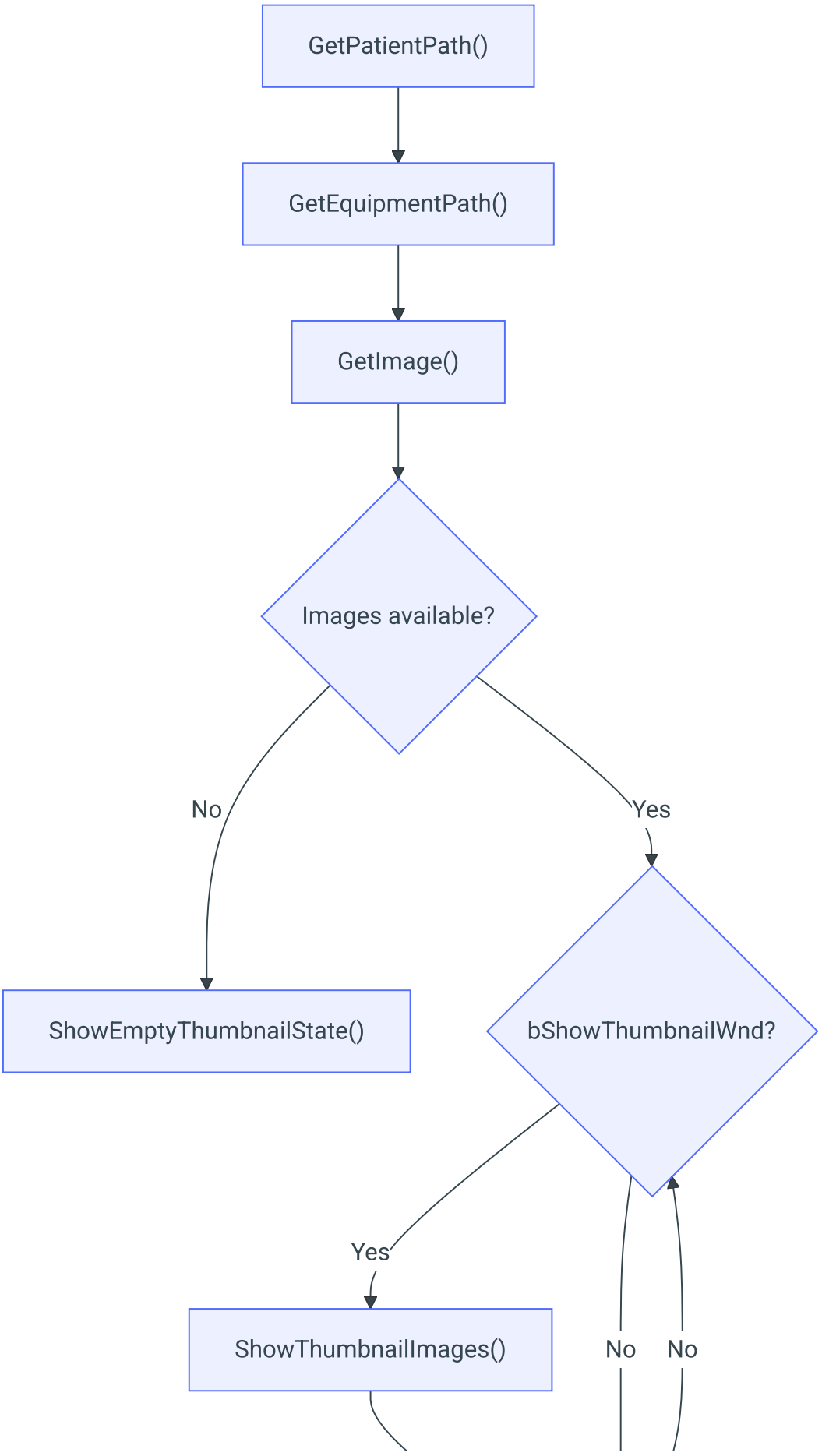
Sub-process: Image Move

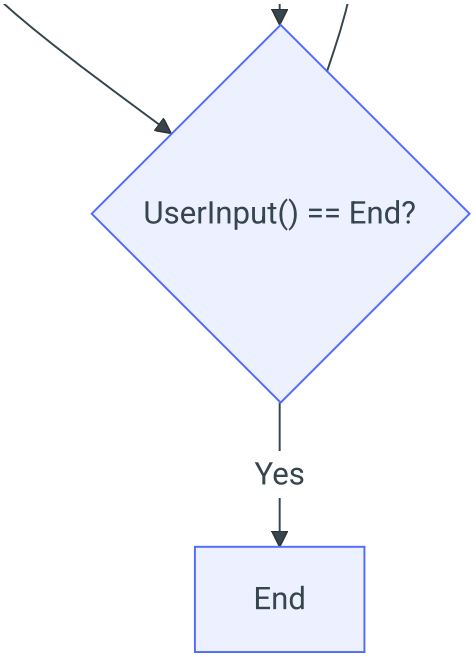
Aspect	Detail
Input	Image and X, Y position value
Output	Changed image position property



SDS-004: Thumbnail Viewer

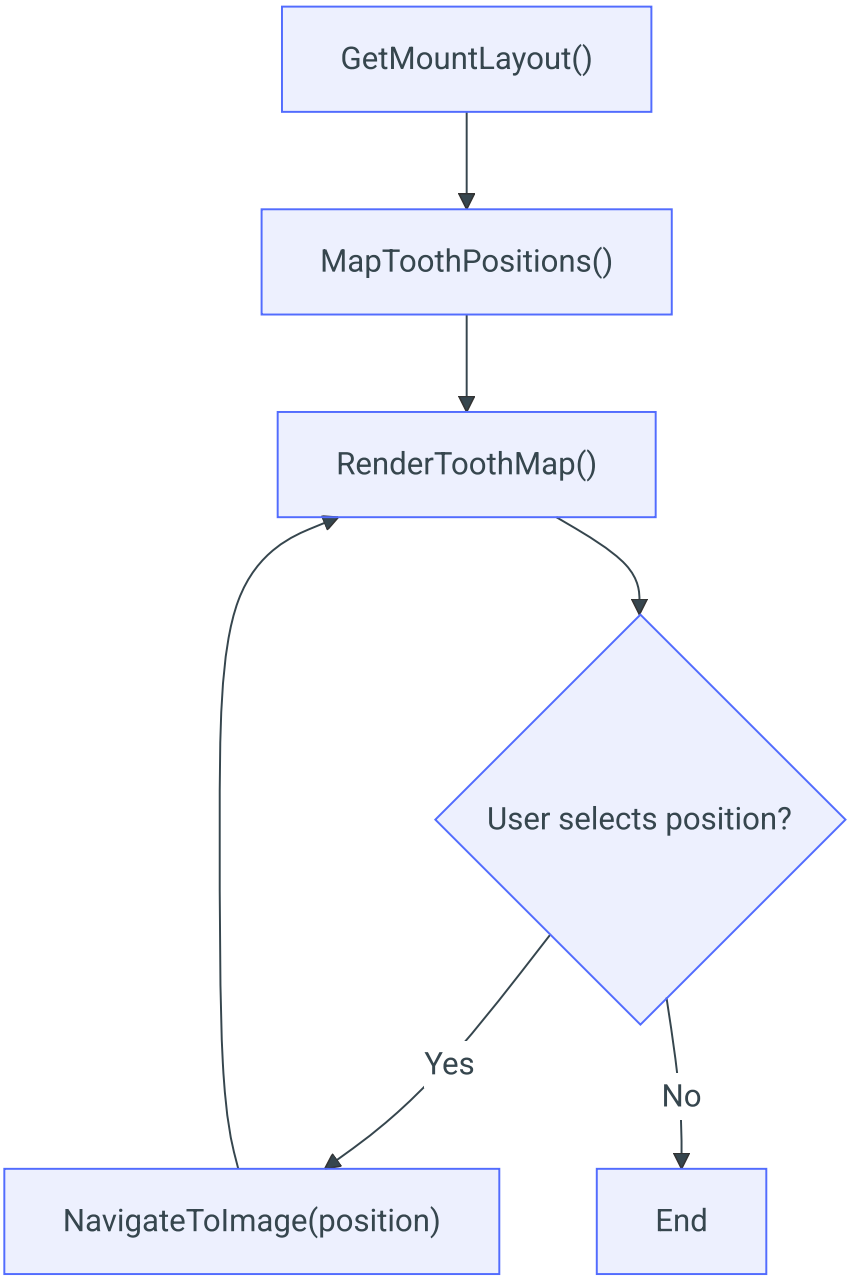
Aspect	Detail
Input	Patient image path
Output	Displayed thumbnail images
Error handling	No images available for selected patient


GetPatientPath()
GetEquipmentPath()
GetImage()
Images available?
No
Yes
ShowEmptyThumbnailState()
bShowThumbnailWnd?
Yes
ShowThumbnailImages()
No
No



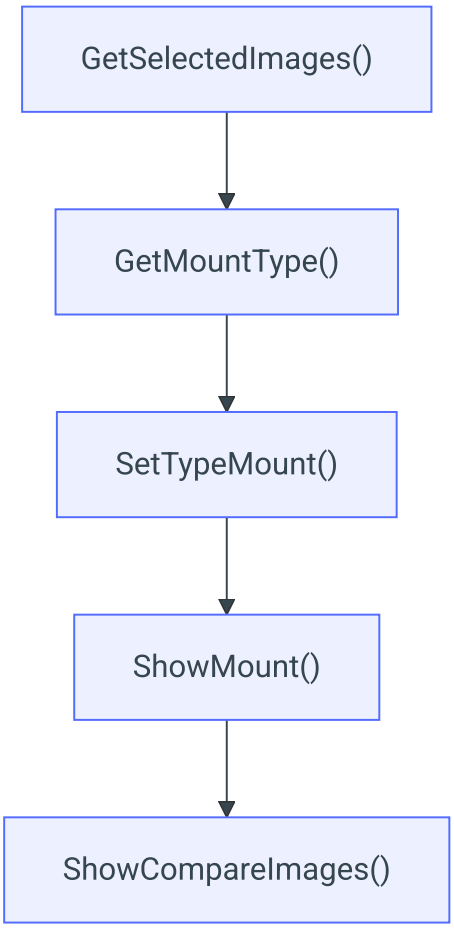
SDS-005: ToothMapWnd

Aspect	Detail
Input	Mounted-study layout and selected exam type
Output	Structured mount or tooth-map view

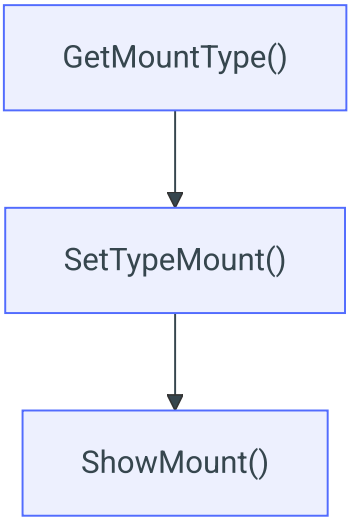


SDS-006: Compare Viewer

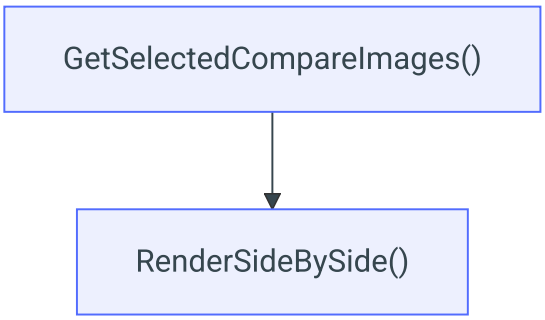
Aspect	Detail
Input	Selected images, user compare action
Output	Compared images display



Sub-process: Show Compare Mount



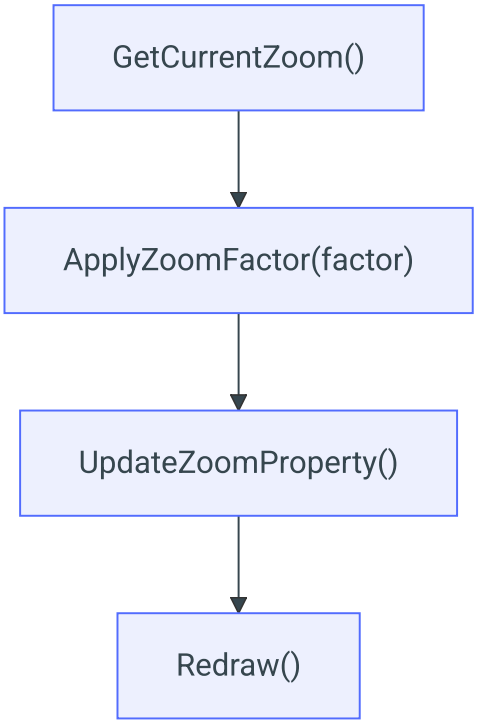
Sub-process: Show Compare Image



5.4 Image Interaction

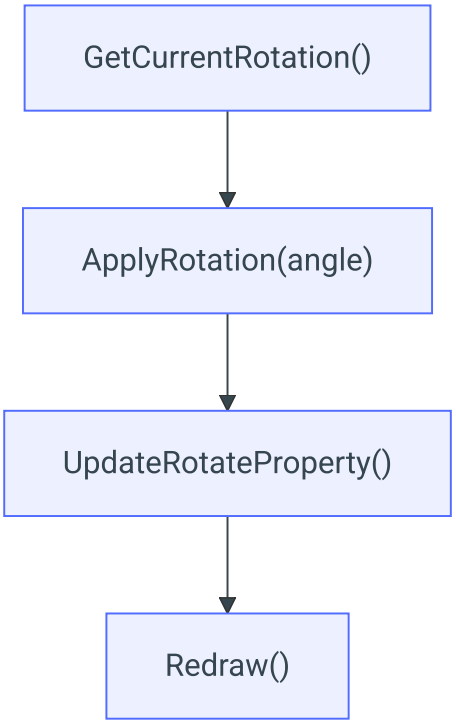
SDS-013: Image Zoom

Aspect	Detail
Input	Image and zoom parameter
Output	Changed image zoom property



SDS-014: Image Rotate

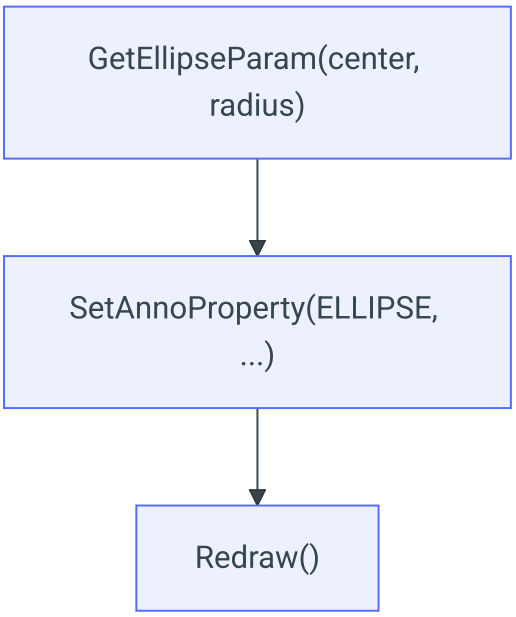
Aspect	Detail
Input	Image and rotate parameter
Output	Changed image rotate property



5.5 Annotation And Measurement

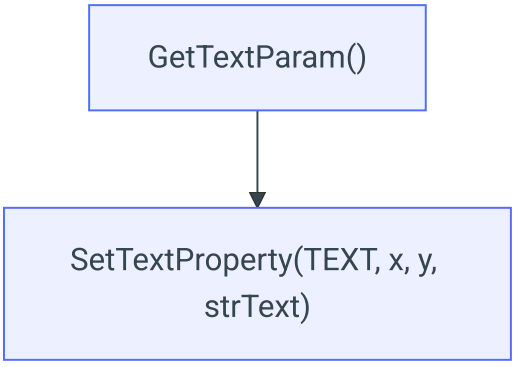
SDS-007: Ellipse

Aspect	Detail
Input	Annotation and ellipse parameter
Output	Changed annotation ellipse property



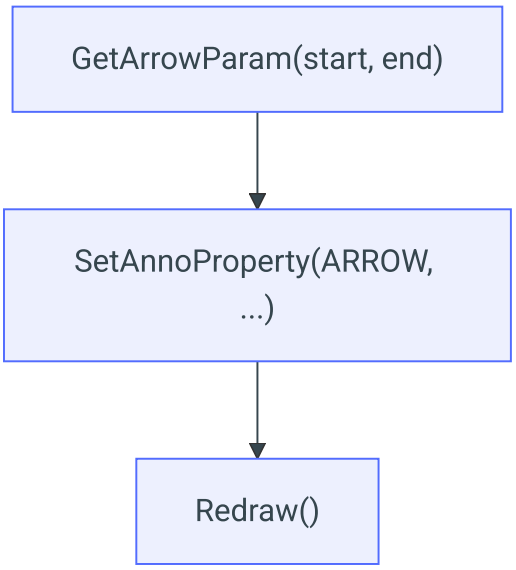
SDS-008: Text

Aspect	Detail
Input	Annotation and text parameter
Output	Changed annotation text property



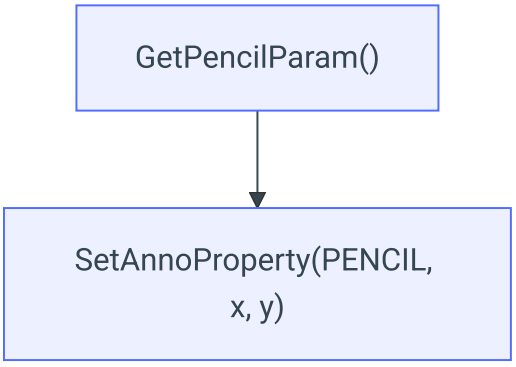
SDS-009: Arrow

Aspect	Detail
Input	Annotation and arrow parameter
Output	Changed annotation arrow property



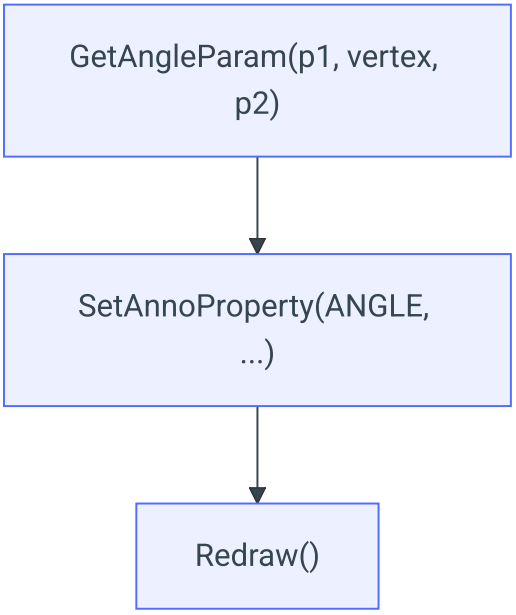
SDS-010: Pencil

Aspect	Detail
Input	Annotation and pencil parameter
Output	Changed annotation pencil property



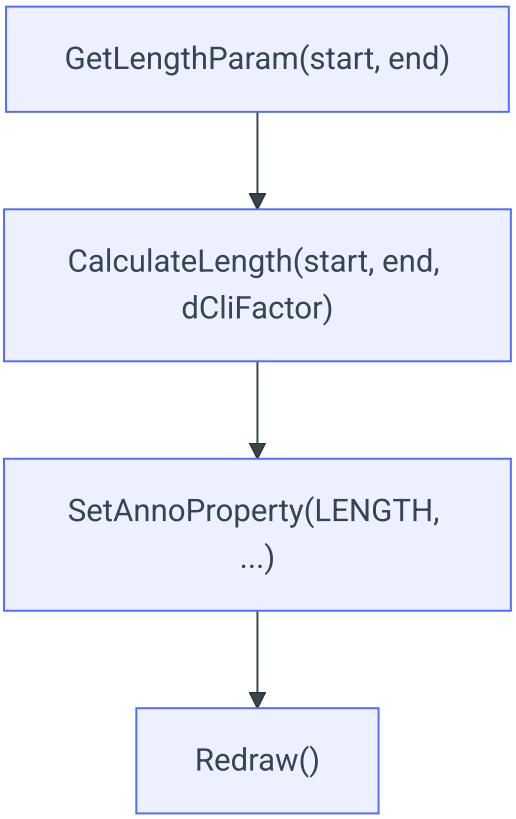
SDS-011: Angle

Aspect	Detail
Input	Annotation and angle parameter
Output	Changed annotation angle property



SDS-012: Length

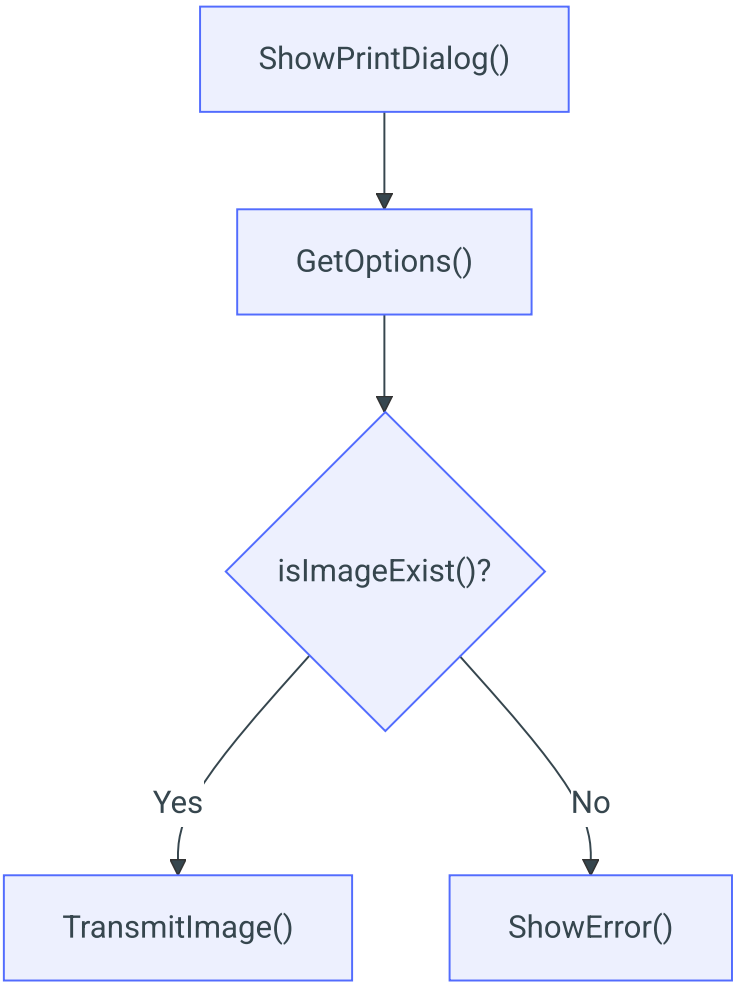
Aspect	Detail
Input	Annotation and length parameter
Output	Calibrated length measurement



5.6 Output And Communication

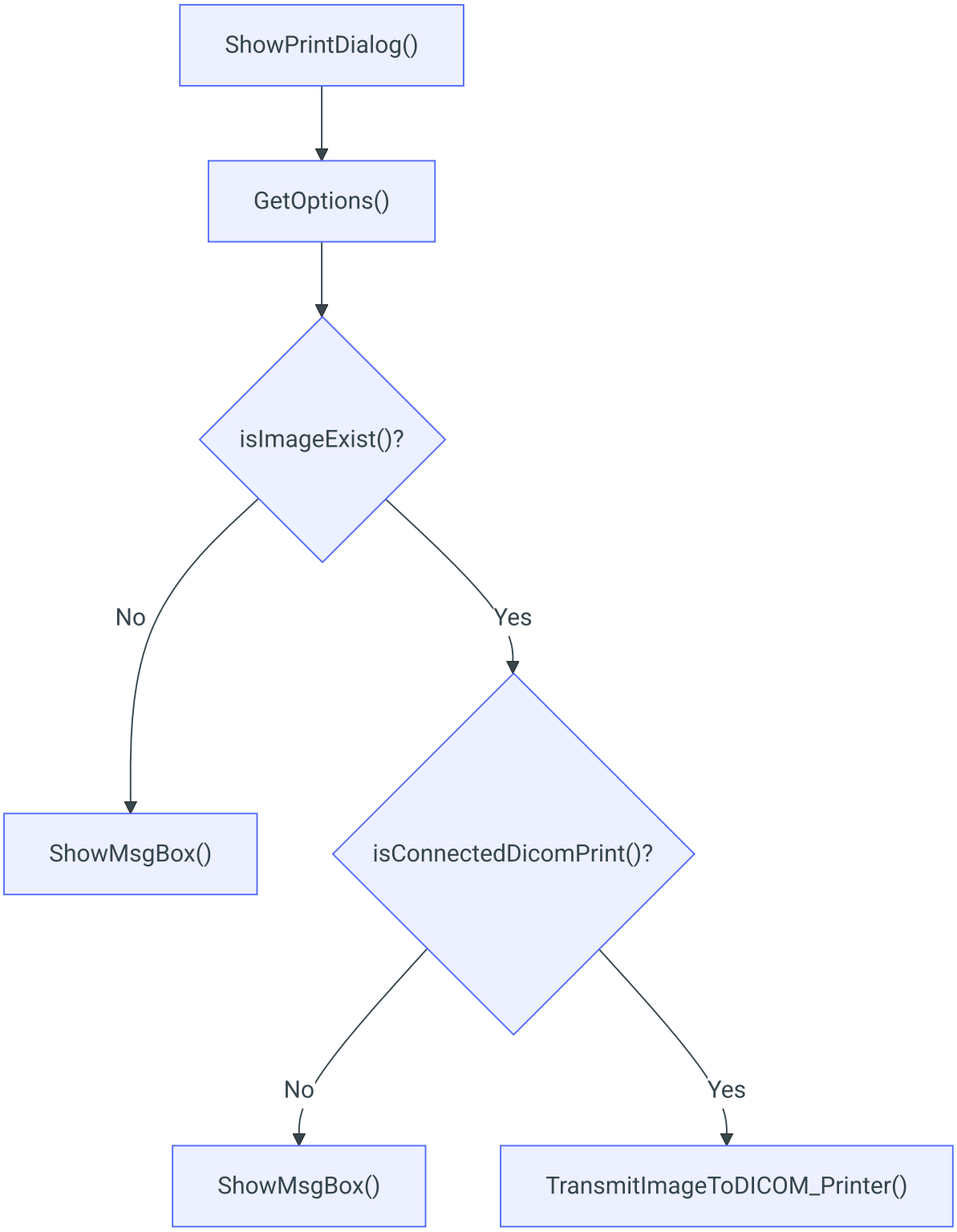
SDS-015: Print Image

Aspect	Detail
Input	Selected image
Output	Print job or error message



SDS-016: Print Image To DICOM Printer

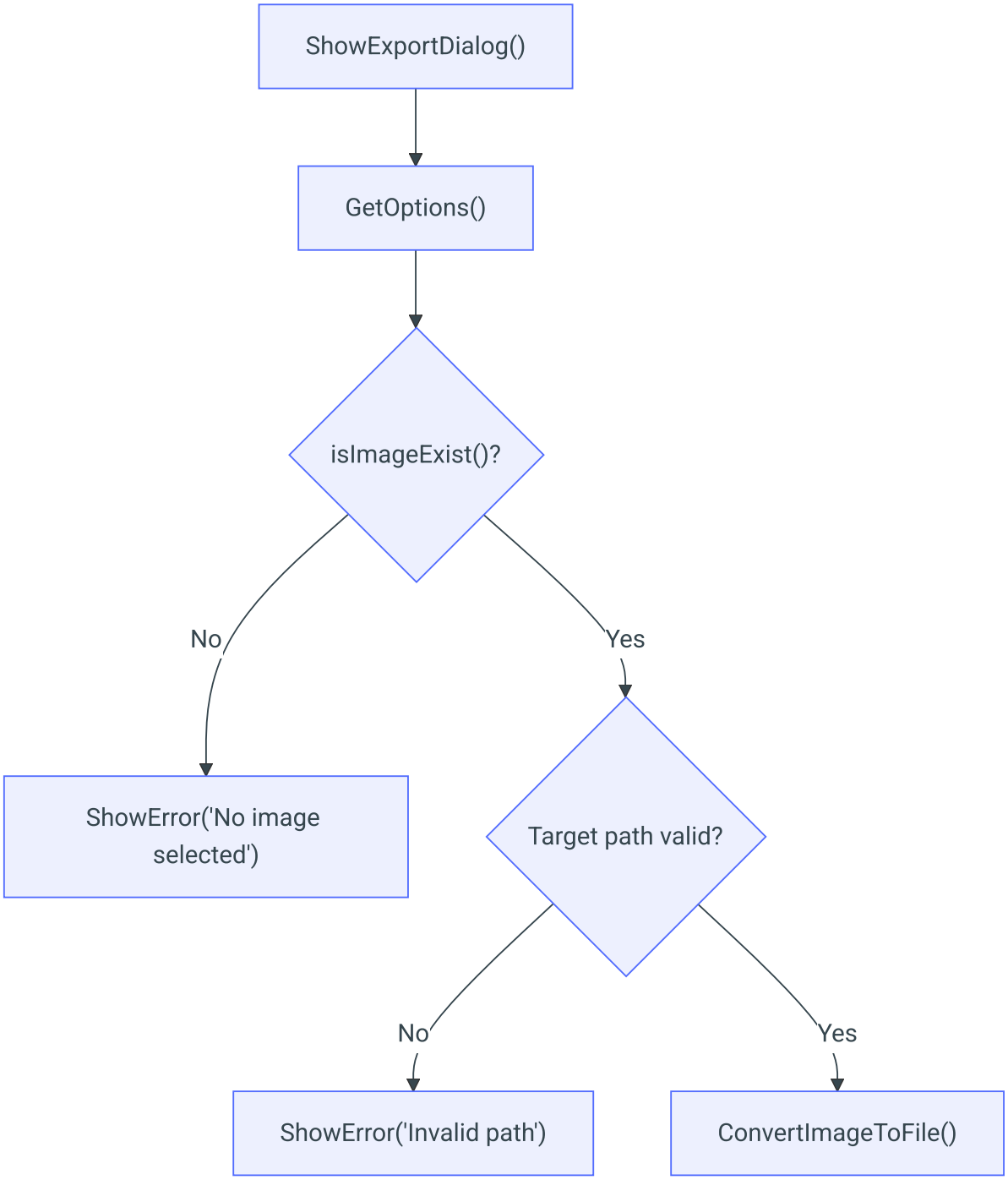
Aspect	Detail
Input	Selected image, DICOM printer configuration
Output	DICOM print request or error message



SDS-017: Export Image To File

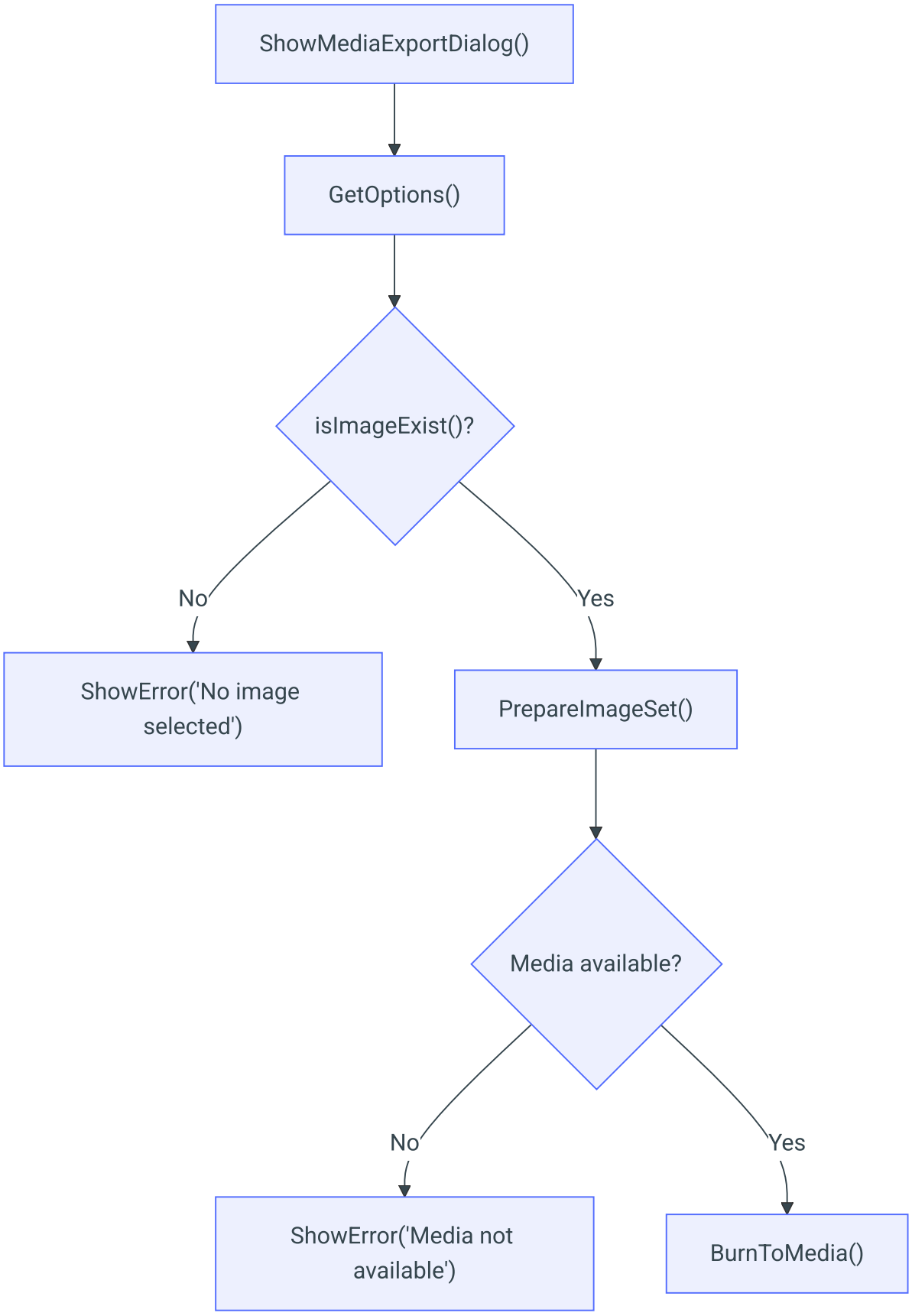
Aspect	Detail
Input	Image

Aspect	Detail
Output	Exported image file
Error handling	No image selected, target path invalid, conversion failure



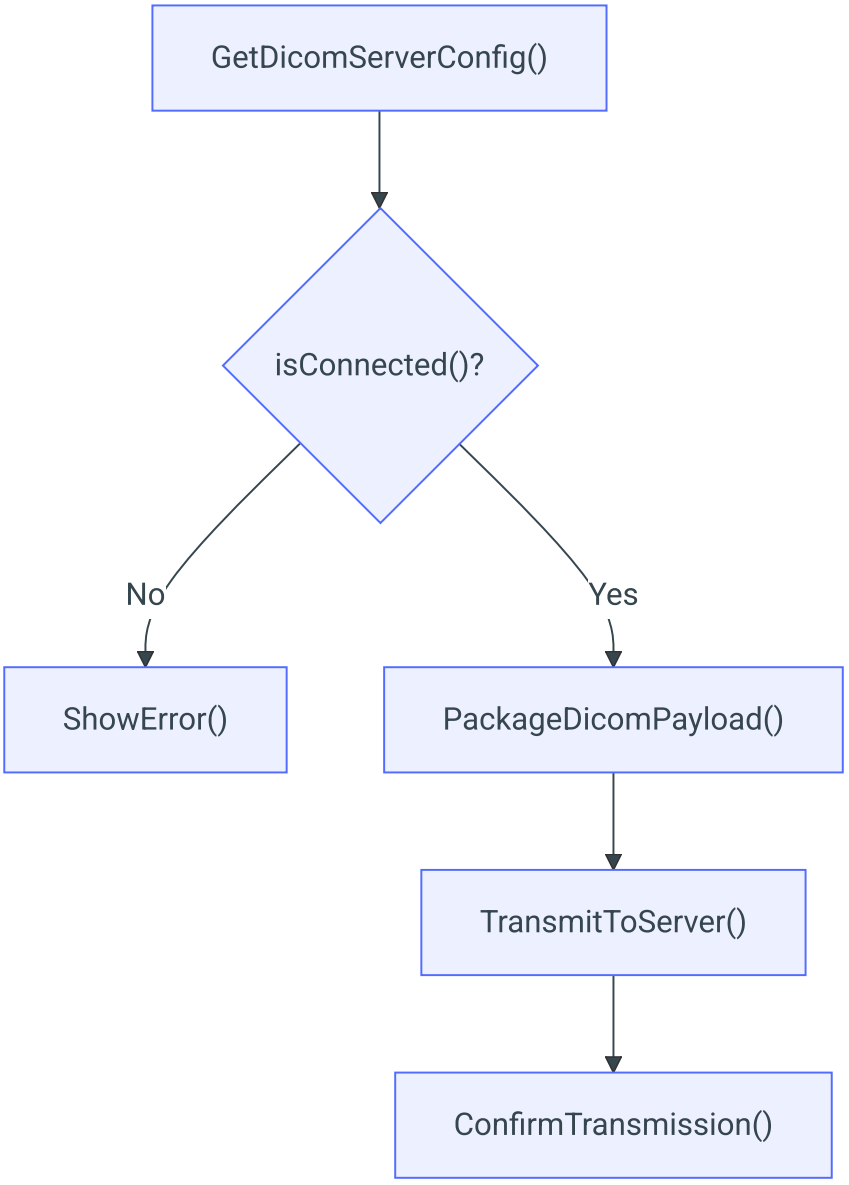
SDS-018: Export Image To CD

Aspect	Detail
Input	Selected image set, media destination
Output	Burned media or error
Error handling	No image selected, media not available, burn failure



SDS-019: SendImageToDicomServer

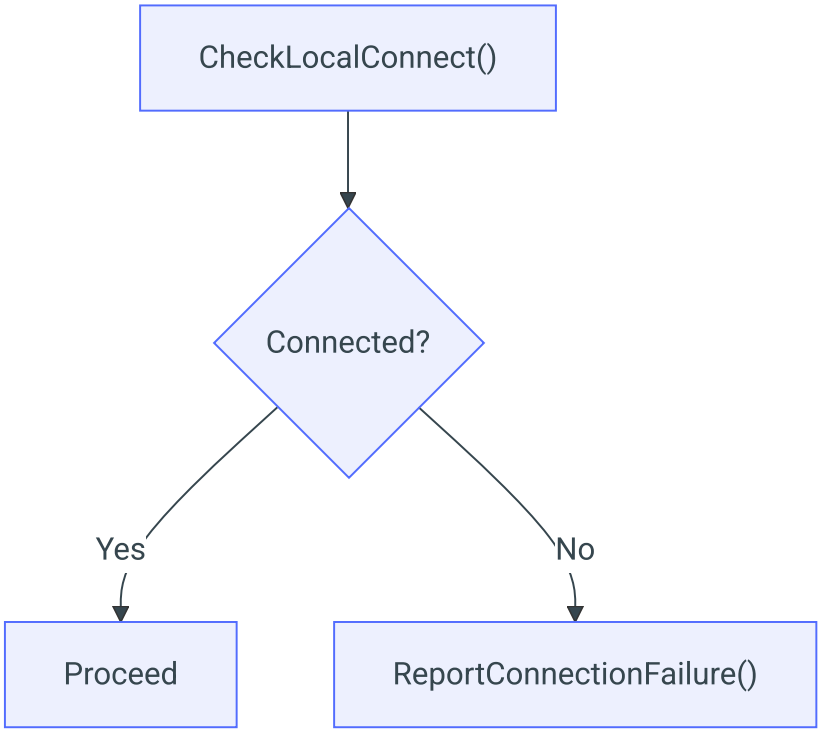
Aspect	Detail
Input	Image
Output	DICOM transmission result



5.7 Support Functions

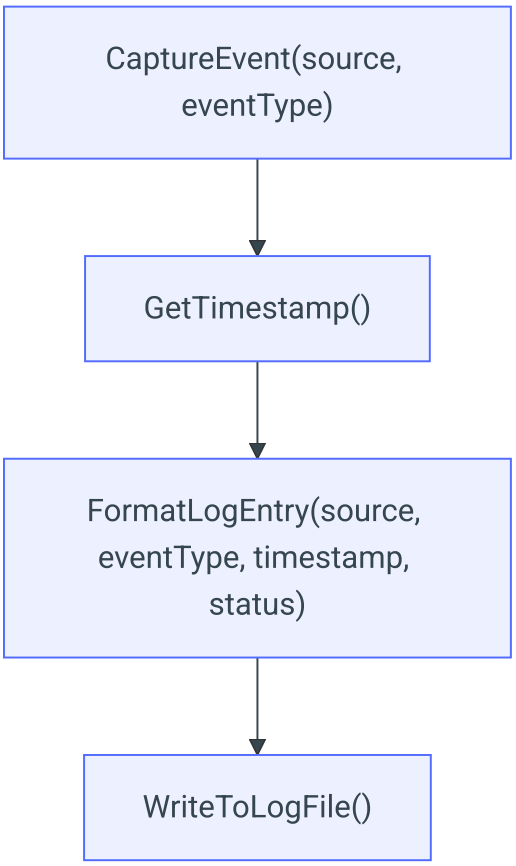
SDS-020: Local Network Connection

Aspect	Detail
Input	Local connection state, device reachability
Output	Verified connection state or failure report



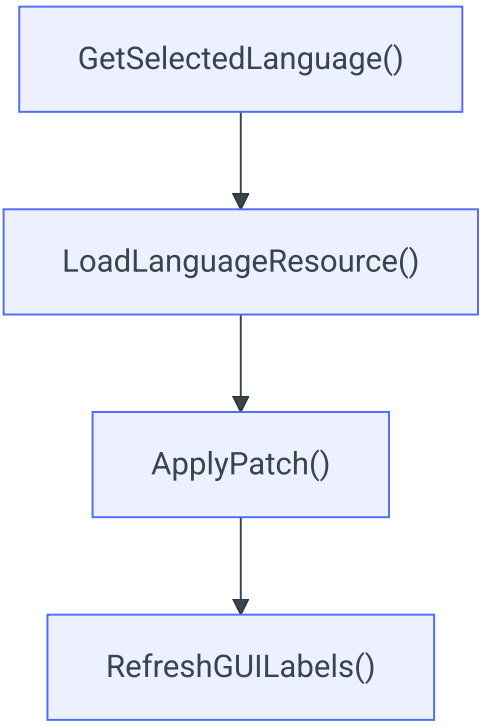
SDS-021: Logging Activity

Aspect	Detail
Input	Any operations, workflow transitions, error conditions
Output	Audit-relevant log records



SDS-022: Patch Language File

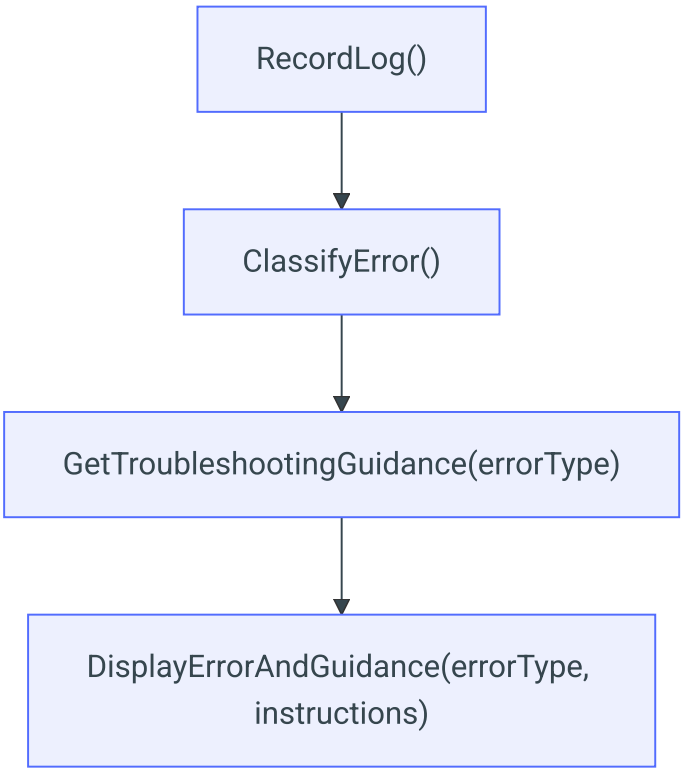
Aspect	Detail
Input	Selected language resource, patch request
Output	Updated GUI language display



SDS-031: Check For Instability

Aspect	Detail
Input	Any operations
Output	Error type specification and troubleshooting instructions

When an error occurs, the system must specify the type of error and provide a solution. Appropriate instructions or guidelines must be provided in case of system failure.



6. Interface And Platform Assumptions

The design assumes:

- a supported Windows workstation environment
- supported acquisition-device connectivity
- file and communication interfaces for export, print, and DICOM exchange
- persistent local storage for patient and study data

6.1 External Interface Contracts

Interface Type	Design Expectation
Sensor and modality interface	Device service must confirm connectivity and acquisition readiness before image display is attempted
File import interface	Imported files must be normalized into the patient-study context before viewer reconstruction occurs
DICOM print interface	Output module must check image availability and print options before transmission

Interface Type	Design Expectation
DICOM server interface	Output module must package and send the selected image set using the configured communication path
Local storage interface	Database, `TMI`, and `TII` data must remain mutually consistent so reopened studies reconstruct correctly
Language-resource interface	GUI text resources must be patchable without changing patient data or image content

6.2 Failure-Handling Expectations

- Selected image missing at print or export time
- Device not found or driver unavailable
- Patient record cannot be reopened from storage
- Local or network connection state prevents intended operation
- System instability requires a logged event and troubleshooting guidance

7. Design Verification

The detailed design described in this document is verified through:

- **Code review**: Completed before unit-level verification as defined in `PV-SV-04` Section 5.2
- **Unit verification**: Each design item is verified through the corresponding unit test procedure in `PV-STP-01`
- **Traceability verification**: Design-to-requirement and design-to-test linkage is maintained through `PV-TM-01`
- **Design walkthrough**: Architecture and inter-unit interface definitions are reviewed as part of the lifecycle review gates defined in `PV-SV-02` Section 5.1

8. Design Use In The SV Set

This design document is intended to support:

- architecture and module allocation in `PV-SV-03`
- validation-basis discussion in `PV-SV-01`
- traceability linkage between requirements, design, and verification in `PV-TM-01`